

QUESTION BANK CAE-3

Subject:[Operating Systems]

Class: SYAI[A] (UCAIL202)

Blooms Taxonomy level

1: Remember, 2: Understand, 3: Apply, 4: Analyse, 5: Evaluate, 6: Create

Q. No.	Question	Marks	BL	CO	PO
UNIT I					
1	Explain time sharing operating system.	4	2	1	1
2	Explain Real time OS with the help of a diagram. List its type.	4	2	1	1
3	Describe the evolution of the operating system.	4	2	1	1
4	What is a multiprocessor system ? Give two advantages to it.	3	1	1	1
5	What is a system call ? Enlist any four system calls.	3	1	1	1
6	Describe six services provided by an operating system.	3	2	1	1
7	Explain layered and microkernel operating system structure.	4	2	1	2
8	Describe the purpose of system calls with Explain how parameter passing is done while implementing system calls.	4	2	1	2
9	Discuss the Basic hardware support necessary for modern operating systems.	4	2	1	5
10	Describe following operating system structures. i) Monolithic ii) Microkernel.	4	2	1	1

UNIT II							
11	What is the process ? Explain the different process state with the help of a state diagram.	4	1	2	1		
12	What is multithreading ? Explain with a suitable diagram.	3	1	2	2		
13	State and describe types of schedules. Describe how each of them schedule the job.	4	2	3	2		
14	Differentiate between short term and long term scheduling.	4	3	3	1		
15	What is pre-emptive and non preemptive scheduling?	4	1	3	2		
16	Calculate average waiting time for following data using First Come First Served (FCFS) and Shortest Job First (SJF) algorithms	4	3	3	2		
	Process					Burst Time	Arrival Time
	P0					8	0
	P1					4	1
	P2					5	2
	P3					3	3
17	Describe one-to-one multithreading model with suitable diagram. Also write any two advantages of one-to-one model over many-to-one model.	4	2	2	2		
18	What is Scheduling? State and explain Three scheduling criteria.	4	1	3	2		
19	Describe I/O Burst and CPU burst cycle. Give one example.	4	2	3	2		
20	Differentiate the process and thread on four points.	4	3	2	2		
UNIT V							
21	Explain the Memory management techniques with its types.	3	2	4	1		

22	Explain techniques of managing memory using first fit, best fit and worst fit with a suitable example.	3	2	4	1
23	Explain swapping and virtual memory systems with examples.	4	2	4	1
24	Describe Demand paging with its advantages.	4	2	4	1
25	List different page replacement policies and explain FIFO, LRU & OPT in details with example.	6	2	4	1
26	Explain Paging with hardware support required for paging	4	2	4	1
27	Explain the concept of segmentation and its advantages.	3	2	4	1
28	Consider the page reference string 2, 3, 2, 1, 5, 2, 4, 5, 3, 2, 5, 2. Calculate page fault & hit ratio for : FIFO, LRU & OPT. (frame size-3).	6	4	4	2
29	Elaborate Contiguous allocation and Index allocation in File system.	6	4	4	2

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